Vibe Shuffle: Relative Valence–Arousal Adaptation for Music Selection

Markus Baumann
Ludwig-Maximilians-Universität München
Munich, Germany
markus.baumann@campus.lmu.de

Felix Hajuj
Ludwig-Maximilians-Universität München
Munich, Germany

Miriam Janner
Ludwig-Maximilians-Universität München
Munich, Germany

Abstract

Music recommenders are good at learning durable preferences, yet the song that fits a listener can change from one moment to the next. We present *Vibe Shuffle*, a browser-based research application that uses lightweight camera and cardiac signals to adapt the next-song choice. The system does not attempt to identify a listener's complete or "true" emotion. After an initialization track, it asks how recent observations differ from that listener's own short reference period. Facial cues provide relative Valence evidence; heart rate and beat-to-beat variability provide relative Arousal evidence. During playback, these estimates form a trajectory in a two-dimensional Valence–Arousal space. The average trajectory point determines one of four broad regions, and the adaptive policy selects the nearest unplayed song in the corresponding Valence–Energy region of a fixed 100-track catalog. The rule is inspectable in the software, and a deliberate local export records the summarized target, region match, song distance, derived signals, and ratings. It does not retain raw camera frames, Bluetooth packets, personal facial-reference values, or the full point-by-point trajectory. An exploratory 15-participant crossover produced a modest but uncertain mood-fit difference for Vibe over Random selection (+0.56 on a seven-point scale, 95% CI $[-0.21, 1.33]$), while the liking point estimate was near zero but imprecise. These data do not establish superiority. They illustrate how an understandable, baseline-relative system can connect multimodal sensing to a testable music-selection decision without presenting proxy signals as an emotion diagnosis.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems.*

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, transparent adaptation

1 Introduction

The next song is a small decision with an immediate effect. A track can support exercise, soften a quiet break, prolong excitement, or feel completely wrong for the moment. People regularly use music to regulate mood and activation [10, 24], but most recommender systems are optimized around comparatively stable information: listening history, favorite artists, similar users, or item features. Those signals describe what a person usually likes. They need not describe what fits now, a broader contextual challenge for music recommendation [11, 25].

Vibe Shuffle began with a practical goal: use signals already available in a browser and an optional wearable to make the next-song choice more responsive to the listener's recent change. The goal was *not* to recover a complete emotional state. Absolute affect recognition would require strong assumptions about facial behavior, physiology, context, and ground truth. A camera observes movement rather than inner experience, and cardiac timing varies with respiration, posture, activity, medication, and fitness. Treating either channel as mind reading would make the interface easy to describe but scientifically misleading.

The project therefore uses a deliberately narrower representation. During setup, the listener supplies short personal references. Subsequent facial and cardiac observations are interpreted as movements relative to those references. The horizontal direction represents more negative or more positive Valence evidence; the vertical direction represents lower or higher Arousal evidence. The point is then placed in one of four broad regions: Tense, Energetic, Melancholic, or Calm. These labels summarize a recommendation region. They are not diagnostic claims about the person. The first track is an initialization exception: it is chosen before a completed listening trajectory, and only later choices can use the preceding track's summarized path.

Songs are organized with the same visual language. Song Valence forms the horizontal axis and song Energy is used as the catalog-side counterpart of listener Arousal. Once Vibe Shuffle has summarized the listener's recent trajectory, it keeps unplayed songs from the matching region and chooses the nearest one. A Random condition selects from the same remaining catalog without using the trajectory. The adaptive step is consequently easy to inspect: *relative signals → recent point → region → nearest available song.*

This paper introduces that software concept and its implementation. It makes four contributions:

- a baseline-relative framing that avoids presenting the application as an absolute emotion detector;
- a transparent multimodal path from camera and cardiac observations to a summarized listener trajectory;
- an inspectable four-region, nearest-song policy over a fixed 100-track catalog; and
- an open browser artifact whose participant view can hide condition labels, with a local row-level export for studying whether this form of adaptation is useful.

An exploratory crossover with 15 participants is retained as an early check, not as the paper's central identity. The observed mood-fit difference favored Vibe on average, but its confidence interval included no difference. The result therefore motivates a better instrumented study rather than a claim that the software already improves experience. The stronger contribution is the explicit bridge between uncertain personal signals and a contestable recommendation decision.

2 Background

2.1 A relative valence–arousal coordinate

The circumplex model represents affect along two continuous dimensions: *valence*, from unpleasant to pleasant, and *arousal*, from low to high activation [21, 23]. Thayer's account of energetic and tense arousal offers a complementary vocabulary for changes in activation and mood [27]. These models are useful here as a compact interaction space, not as a complete account of emotion. Both dimensional and discrete models appear in music-emotion research, with results sensitive to model and annotations [6].

Figure 1 is an independently drawn conceptual synthesis: it places the continuous and categorical affect vocabularies beside the strictly group-level HRV-frequency context that motivates an exploratory diagnostic signal.

Vibe Shuffle borrows only the two-axis coordinate structure. It does not output the emotion words shown in Figure 1 or treat them as ground-truth labels. The coherence profiles in panel (c) are likewise not the prototype's diagnostic signal.

Vibe Shuffle places the listener at a point (V, A) in this space and divides the plane into four quadrants. Crucially, the point describes change relative to that listener's own reference measurements. The center is the personal baseline; evidence of more positive facial display moves the point right, while evidence of greater cardiac activation moves it upward. The system therefore asks a deliberately narrow question—which direction did the observed signals move?—rather than attempting to name or recover the listener's full emotional state. This distinction matters because facial movements are not unique fingerprints of specific emotions and vary with person and context [4]. Facial judgments can also shift with body context, and Valence–Arousal relations vary across people and cultures [1, 13].

2.2 Connecting listener state to music

Music can be placed in a parallel two-dimensional space. Song *Valence* describes conveyed positivity, whereas *Energy* describes perceived intensity and activity. Prior emotion-oriented music interfaces have used these attributes as musical counterparts to affective valence and arousal [8]. Vibe Shuffle adopts that practical bridge: listener Valence is compared with song Valence, and listener Arousal is compared with song Energy. The alignment is an engineering assumption, not a claim that musical Energy and physiological Arousal are the same construct. Music-emotion recognition also uses continuous coordinates, but its annotations remain subjective [29].

The shared plane makes the selection rule easy to inspect. High and low values on each axis produce four broad regions, and the listener's relative position determines the target region. Within that region, distance in the continuous plane distinguishes closer from more distant songs. The quadrants simplify communication; the underlying coordinate preserves degree and direction.

2.3 Heart-rate variability as relative evidence

Heart-rate variability (HRV) describes variation between successive heart beats. The root mean square of successive differences (RMSSD) is a widely used time-domain summary of short-term RR-interval variation [26]. Heart rate and RMSSD depend on respiration, movement, posture, fitness, medication, sensor quality, and other factors; neither measure identifies a particular emotion on its own. Good-practice guidance therefore emphasizes standardized conditions, artifact review, matched durations, and respiration assessment [14].

Vibe Shuffle consequently uses cardiac measurements only as baseline-relative evidence for the arousal axis. A heart rate above the listener's reference and an RMSSD below that reference contribute upward evidence; changes in the opposite direction lower it. The mapping is transparent and testable, but remains a prototype heuristic rather than a clinical or psychological measurement.

3 Related Work

3.1 Emotion-aware music selection

Music can evoke, express, maintain, or help regulate affect through several mechanisms [10]. Emotion-aware music systems turn that relationship into an interaction: the listener supplies or reveals context, and the system uses it to narrow the next musical choice.

Feel the Moosic is a close design reference. Helmholz et al. simplified the Russell and Thayer representations into color-coded interfaces in which a listener explicitly selected a position; the prototype then requested songs within corresponding Spotify Valence and Energy ranges [8]. Vibe Shuffle retains the legible two-axis mapping but changes the interaction contract. It estimates only baseline-relative movement from a camera and an optional cardiac sensor, summarizes a trajectory over the listening window, and selects an unplayed catalog track near that point. This is an implicit and auditable adjustment mechanism, not a stronger claim to emotion recognition.

3.2 Personal and multimodal affective recommenders

Several systems show that personal physiological or camera cues can inform music selection. Janssen et al. learned listener-specific relations between music and physiological response and used them to steer an affective music player toward a goal [9]. Ayata et al. compared machine-learning models that predicted Valence and Arousal from wearable galvanic-skin-response and photoplethysmography measurements before recommending music [2]. Tran et al. used a facial image to estimate a point in Valence–Arousal space and retrieve nearby songs in Spotify feature space [28].

These projects establish the feasibility of connecting affective signals to a music interface. They also clarify Vibe Shuffle's narrower contribution. Vibe does not optimize classification accuracy, learn a high-capacity personal model, or claim a target emotional

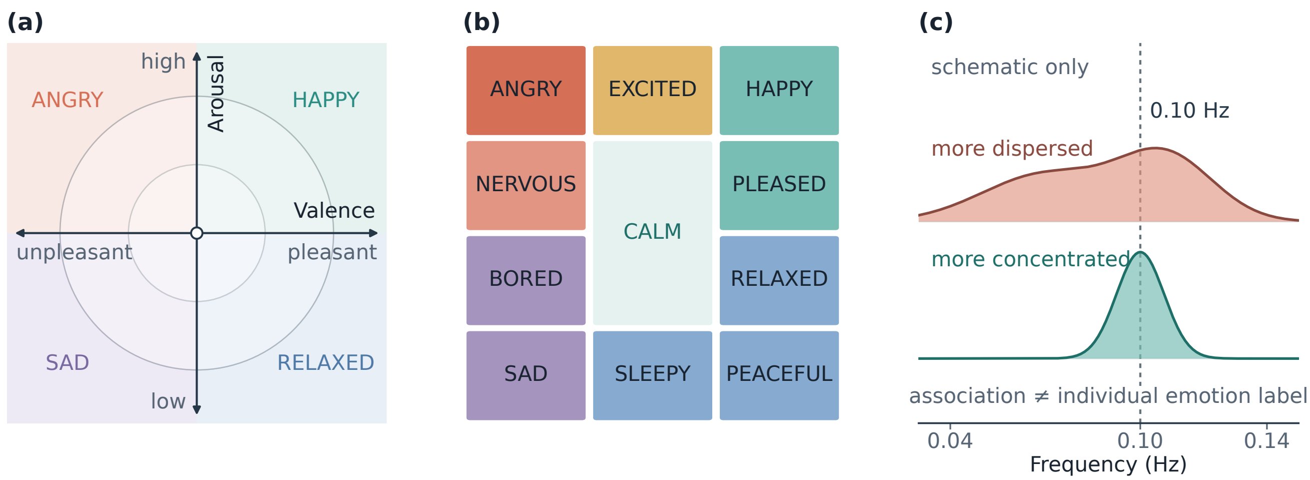

Figure 1: Conceptual foundations. Panels (a)–(b) are adapted and independently redrawn from Helmholz et al. [8], Figure 1, which operationalizes Russell's circumplex [23] and Thayer's mood–arousal model [27]; no source pixels are reused. Panel (c) is an original, data-free schematic of the group-level frequency distinction discussed by Balaji et al. [3]; no curves, observations, or source artwork are reproduced or adapted. Illustrations only—not Vibe measurements or classifiers.

state. It makes a small rule-based decision from recent baseline-relative evidence and exposes the summarized target, region, and distance. Recommendation utility is measured separately from the plausibility of the signal label; the export does not preserve every trajectory sample.

3.3 HRV, coherence, and the boundary of inference

HRV provides useful information about cardiac dynamics, but its interpretation depends on measurement duration and context [26]. Balaji et al. analyzed 1,884,216 mobile HRV-biofeedback sessions from 70,041 users and reported aggregate associations between coherence measures and emotional labels entered by users after their sessions [3]. Sessions labeled with positive emotions had higher mean coherence and more stable coherence frequencies than sessions labeled with negative emotions. The study was an observational analysis of 3–15 minute biofeedback practice, lacked pre-session emotion reports, and did not establish a per-window emotion classifier.

That boundary directly shapes Vibe Shuffle. The software exports a custom normalized spectral-peakedness proxy for later diagnostic analysis. It is not the coherence measure analyzed by Balaji et al., and it does not determine the listener coordinate, quadrant, or next song. Selection uses the baseline-relative heart-rate and RMSSD arousal estimate instead. Thus, the coherence study motivates a cardiac-rhythm feature worth observing; it neither validates this proxy nor justifies inferring a listener's emotion from it.

3.4 Transparent and user-centered recommendation

Recommender explanations can target transparency, trust, effectiveness, or decision support [20]. Vibe exposes target, region, eligibility, and distance rather than adding a post-hoc narrative. Inspectability alone does not ensure experience: perceived quality, effort, privacy, and context still matter [12].

3.5 Positioning Vibe Shuffle

Vibe Shuffle brings three ideas together: the understandable two-axis music map, personal rather than population-relative sensing, and a controlled comparison with Random selection from the same catalog. Its purpose is not to advance a new emotion classifier. It is to make the complete path from an uncertain observation to a recommendation visible to the listener, simple enough to inspect and specific enough to test.

4 From Relative Signals to the Next Song

4.1 Design principle

Vibe Shuffle is a client-only React/Vite application. A camera, an optional Bluetooth Heart Rate Service device, and Spotify playback meet inside one browser session. There is no study database or recommendation server. The design keeps adaptation between songs rather than changing playback continuously: the system listens to the latest window, summarizes it, and then chooses the next track. This makes the decision legible and prevents every short-lived fluctuation from becoming a command.

Figure 2 shows the complete idea. Two personal references anchor the incoming signals. Confirmed playback produces a path rather than a single instantaneous label. The path is averaged, mapped to a region, and used to select a nearby song. A participant then rates the track, closing the interaction loop.

4.2 Relative Valence from camera cues

The browser runs MediaPipe Face Landmarker locally [16]. It returns facial blendshape activations such as smile, cheek, brow, eye,

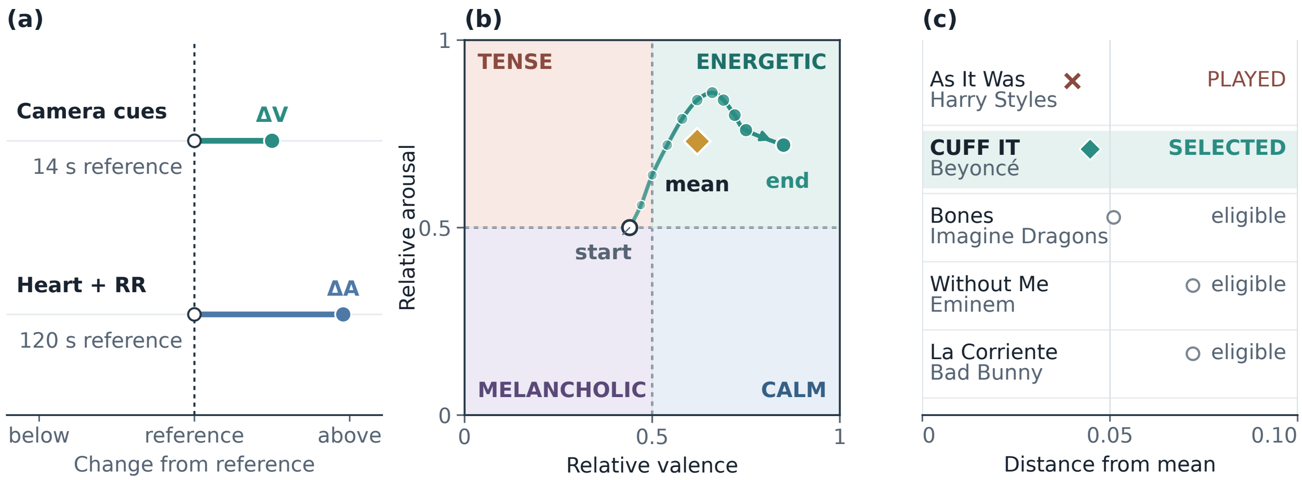

Figure 2: Relative sensing, trajectory, and selection in Vibe Shuffle. (a) Camera and cardiac inputs are expressed as signed changes from personal references. (b) A curved Valence–Arousal trajectory is summarized by its arithmetic mean (gold diamond); points grow and darken from the open start marker to the filled endpoint. (c) Within the matching region, *As It Was* is nearest but excluded as played (cross), so *CUFF IT* becomes the selected eligible track (filled diamond); open circles are alternatives. The relative changes and trajectory are illustrative; catalog candidates, distances, and the selection result are repository-derived. This is a relative heuristic, not an emotion diagnosis; coherence is logged separately and does not affect selection.

jaw, frown, and mouth movement. Vibe Shuffle combines a selected subset into transparent positive, negative, and tension-related scores. These labels belong to Vibe Shuffle's rule layer; MediaPipe itself does not output the project's emotion-region labels.

A 14-second calibration captures a short facial reference, and a slow running baseline continues to adapt. Later observations are debiased against this reference and smoothed across frames. Positive facial evidence moves Valence to the right; negative or tension-related evidence moves it to the left. If no face is detected, Valence returns to the neutral center. The current heuristic also lets visible movement increase positive Valence evidence and upward Arousal evidence. Movement may instead reflect dancing, repositioning, lighting change, or camera noise, so this rule is not a validated engagement or affect measure.

4.3 Relative Arousal from cardiac timing

The optional wearable provides device-produced beats per minute and, when available, RR intervals—the elapsed time between successive beats. The application does not receive a raw electrocardiogram and cannot inspect the device's peak-detection process. Implausible and abruptly changing RR intervals are rejected before short-term variability is calculated.

The primary variability feature is the root mean square of successive differences (RMSSD). A 120-second reference stores the listener's ordinary median heart rate and typical RMSSD together with robust estimates of their variation. During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward Arousal evidence. Heart rate receives the larger influence because RMSSD estimated from a short window is unstable; when the two disagree, the RMSSD contribution is strongly reduced. At least 20 accepted RR intervals are required before the full estimate is treated as usable. Controlled resting data show high agreement between 120-second and longer RMSSD recordings [18]; this supports the calibration duration, not short moving-wearable windows or Arousal validity.

The implementation also exports SDNN and a custom normalized frequency-domain spectral-peakedness score. The CSV calls this the physiology coherence field. It is an experimental diagnostic, not the coherence measure analyzed by Balaji et al., and it does not determine the selected region. Prior HRV-biofeedback research reports group-level associations between coherence patterns and end-of-session self-reported emotion categories [3]. That result motivates careful study of cardiac timing, but it neither validates the prototype's proxy nor shows that a short personal window can identify an individual's emotion. Vibe Shuffle accordingly uses baseline-relative activation evidence rather than translating coherence frequencies into labels.

4.4 A trajectory rather than a snapshot

Whenever the fused live coordinate changes during confirmed playback, Vibe Shuffle appends a point to the current trial buffer. Facial cues primarily control Valence. Usable cardiac evidence supplies the Arousal base, with visible movement able to add upward Arousal. If cardiac evidence is unavailable, camera movement provides an upward-only fallback. If neither channel is usable, the coordinate returns to $(0.5, 0.5)$.

The resulting sequence makes the listener's path visible conceptually:

$$\mathcal{T} = \{(V_1, A_1), \ldots, (V_m, A_m)\}.$$

At rating time, the application uses the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m} \sum_{i=1}^{m} (V_i, A_i)$$

as the target for the next adaptive choice. The points are event-driven rather than sampled at a fixed rate, so this is not a time-weighted physiological average. Sample count and actual listening duration are exported to make that approximation visible.

4.5 Four regions and a fixed music space

Splitting both axes at 0.5 produces four recommendation regions: Energetic (higher Valence and Arousal), Calm (higher Valence and lower Arousal), Tense (lower Valence and higher Arousal), and Melancholic (lower on both axes). The frozen catalog contains 100 Spotify tracks, with 25 tracks in each corresponding Valence–Energy region. Track coordinates originate from historical Spotify features in a public dataset [17]. Energy is a practical catalog-side counterpart of Arousal, not proof that an audio feature and a physiological construct are identical.

Both Random and Vibe exclude the current and all previously played tracks. Random ranks the remaining tracks with a session-seeded pseudorandom score. Vibe first retains available tracks in the target region and then selects the smallest Euclidean distance:

$$s^* = \arg \min_{s \in Q(\bar{V}, \bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

The seed resolves only an exact distance tie. If a region has no remaining track, the policy falls back to the complete unplayed pool. The export records the target coordinate, signal source, distance, and region match. It does not record the random seed, the personal facial-reference values, or individual trajectory points. The first track is also chosen from the live setup estimate before the 14-second facial reference is finalized; later choices use the preceding listening window.

4.6 Browser implementation and privacy boundary

Spotify authentication uses Authorization Code with PKCE, and the Web Playback SDK provides playback for an allowlisted Premium account. Raw camera frames, facial landmarks, and Bluetooth notifications remain in browser memory. The locally downloaded CSV contains derived signal summaries, song metadata, selection fields, ratings, quality flags, pseudonymous identifiers, and timestamps. It is downloaded automatically at protocol completion and can also be downloaded manually. It contains no image, landmark set, raw waveform, or audio. Because it retains only trajectory summaries, it supports review of the final adaptive target but not replay of every sensing step.

This is a bounded privacy property, not a claim of offline operation. Spotify authentication and playback, MediaPipe asset delivery, and GitHub Pages create ordinary external traffic. Exported participant numbers and timestamps are pseudonymous rather than anonymous and still require appropriate research-data handling.

Table 1: Exploratory condition summaries ($N = 15$ participants). Confidence intervals are two-sided. The reported study tests used one-sided Vibe-over-Random alternatives.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]

5 Exploratory Crossover Study

5.1 Protocol

The implemented study asks whether the relative adaptive policy feels more appropriate than Random selection from the same catalog. Every participant completes one five-track Random block and one five-track Vibe block. Two masked protocols reverse the order, and odd/even participant numbers suggest the assignment. Participants see a protocol identifier rather than the condition name when the operator keeps the participant-facing mode enabled. The same interface exposes a view toggle and protocol controls, while the experimenter and exported data can reveal the order. Masking therefore depended on study administration; it was not technically enforced or double blind. The paired design is efficient but remains susceptible to practice and carryover, making reversed order a substantive control [7].

Each track plays for up to 60 seconds. Participants may select *Rate now*; the actual duration and an early-rating flag are preserved. After each track, the interface collects a seven-point liking rating, a seven-point mood-fit rating, and a categorical current-mood report. Mood fit is treated as the primary outcome in this exploratory report because it directly evaluates the matching idea. Liking is treated as secondary and helps separate contextual fit from general song preference.

5.2 Available evidence

The supplied analysis report publishes one five-trial condition mean per participant/session for 15 participants, together with paired tests and condition-level summaries. These 15 rows are frozen in the repository under generic analysis IDs. The protocol implies 75 planned ratings per condition and outcome, but the repository still lacks the underlying trial rows needed to verify completeness, exclusions, or within-session variation. It also lacks recruitment records, demographics, and the complete study-administration file. The present paper therefore treats the participant/session as the inferential unit and does not claim a fully reproducible human-study archive.

Paired differences are defined as Vibe minus Random. We show the published participant/session means, report condition means and standard deviations, paired mean differences with two-sided 95% confidence intervals, and the originally supplied paired tests. Because no dated preregistration was available, all inference is exploratory [19]. We foreground paired effects and confidence intervals rather than a thresholded p-value [5].

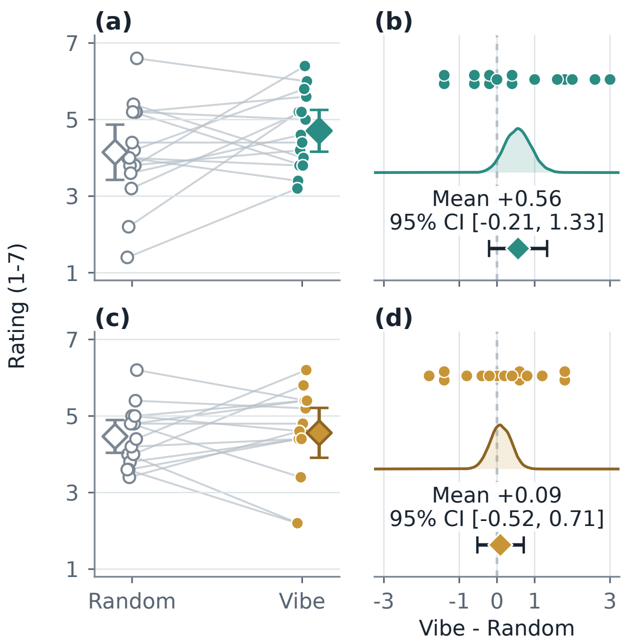

Figure 3: Participant-level paired outcomes. Panels (a)–(b) show mood fit; panels (c)–(d) show liking. Left panels connect Random and Vibe ratings and show condition means with 95% confidence intervals. Right panels show participant differences, bootstrap mean distributions, and paired mean 95% confidence intervals; both intervals cross zero.

6 What the Prototype Shows—and Does Not Show

6.1 A directional but uncertain mood-fit result

Mood fit averaged 4.71 under Vibe and 4.15 under Random (Table 1). The paired mean difference was +0.56 rating points with a 95% confidence interval from −0.21 to 1.33. The supplied directional paired test did not reach $\alpha = .05$, $t(14) = 1.56$, $p = .070$; the signed-rank sensitivity check agreed, $W^+ = 74.0$, $p = .088$. The standardized paired effect was $d_z = 0.40$.

Figure 3 preserves the original paired-estimation view while making the comparison easier to scan. Eight participant/session means were higher under Vibe, one tied, and six were lower. The average is compatible with a useful improvement, but the interval also allows little benefit or a small disadvantage. The correct reading is therefore *promising direction, unresolved magnitude*, not validation of the adaptive policy.

Liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09 with a 95% confidence interval from −0.52 to 0.71, $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. This near-zero point estimate suggests that the mood-fit direction was not simply a broad rise in song liking. It does not establish equivalence: the interval remains wide enough to contain meaningful changes in either direction.

6.2 Artifact validation

The software's scientific core is separated into pure JavaScript modules so it can be checked without a camera, wearable, or Spotify account. The repository tests catalog size and quadrant balance, CSV finiteness and quality flags, facial-rule behavior, RR filtering and baseline-relative cardiac behavior, and Random/Vibe selection. In the verified repository state, all 63 deterministic tests pass. These checks validate internal contracts; they do not validate the facial or physiological interpretation against ground truth.

The fixed catalog and crossover hold many interaction details constant: playback path, rating questions, number of exposures, and available song pool. That makes the selection policy easier to study. It does not make the two conditions acoustically identical, because the policies can choose different artists, genres, languages, familiarity levels, and popularity profiles.

6.3 Measurement and study limits

The principal limitation is construct validity. Facial behavior cannot be reduced reliably to a universal readout of felt emotion [4]. Vibe's rule weights were not learned or validated on a labeled affect dataset. Lighting, pose, occlusion, morphology, culture, and deliberate display can all change the output.

Cardiac timing is similarly non-specific. Respiration, posture, speaking, movement, fitness, medication, contact quality, and vendor processing can alter heart rate and RMSSD [22, 26]. The listening window is short for HRV, respiration is not measured, and the application receives processed RR intervals rather than raw peaks. A personal baseline reduces between-person differences but cannot remove these contextual influences.

Movement can affect the camera channel, raise heart rate, and introduce RR artifacts at the same time. The current fusion can therefore count one behavior more than once. The trajectory mean is event-driven rather than time-weighted, and the first selection may occur before both references are fully established. Optional sensing also means that participants can experience meaningfully different evidence paths.

At the exact no-signal center, current code paths also disagree on which region owns the boundary: the fallback label is Calm, whereas the numerical greater-than-or-equal rule assigns $(0.5, 0.5)$ to Energetic. This edge case should be unified before a confirmatory study.

The human-study archive adds separate constraints. Raw trial rows, participant characteristics, recruitment and compensation details, the exact sensor model, room conditions, consent documentation, and ethics procedure are not present in the repository. These are reporting gaps, not evidence that procedures were absent. They prevent a complete independent audit and rule out a strong effectiveness claim.

6.4 Next validation steps

A stronger study should proceed in stages. First, validate the frozen facial and cardiac mappings under controlled changes in expression, lighting, posture, movement, and respiration. Second, make successful completion of calibration, timing, missing-data flags, and a ten-trial export an end-to-end release gate. Third, preregister a meaningful mood-fit effect, sample size, exclusions, primary paired analysis, order analysis, and sensor-quality checks before collecting a larger crossover sample. Sample size should target interval precision or a smallest relevant effect, not copy the pilot [15].

The interface should also let the listener choose the adaptation goal. The current policy matches the recent region. A future system could instead maintain, intensify, or deliberately shift it. Keeping those goals separate is essential: otherwise a recommendation cannot be judged against a clear intention.

7 Conclusion

Vibe Shuffle demonstrates a simple alternative to absolute emotion recognition. It compares recent camera and cardiac observations with a listener's own reference, follows their movement through Valence–Arousal space, and turns the average trajectory into an understandable next-song decision. The four regions reduce an uncertain continuous estimate to a transparent shortlist; nearest-song selection then makes the final action reviewable at the exported decision-summary level.

The exploratory study does not prove that this policy improves listening. Fifteen participants showed a modest average mood-fit direction and almost no average liking change, but both uncertainty intervals included zero. More importantly, neither that result nor the software establishes an objective emotion label. Vibe Shuffle contributes an inspectable research object: every assumption can be located in a reference period, a signal rule, a trajectory, a region boundary, or a distance, even though the present export does not retain every intermediate sample.

That transparency is the project's strongest foundation. With controlled signal validation, time-aware fusion, complete study provenance, and a larger preregistered comparison, the same artifact can test when relative adaptation helps, for whom it helps, and which signal path contributes. Physiological input becomes useful here not as mind reading, but as one uncertain, personal, and visible influence on a familiar choice: what to play next.

References

[1] Hillel Aviezer, Yaacov Trope, and Alexander Todorov. 2012. Body Cues, Not Facial Expressions, Discriminate Between Intense Positive and Negative Emotions. *Science* 338, 6111 (2012), 1225–1229. doi:10.1126/science.1224313

[2] Değer Ayata, Yusuf Yaslan, and Mustafa E. Kamasak. 2018. Emotion Based Music Recommendation System Using Wearable Physiological Sensors. *IEEE Transactions on Consumer Electronics* 64, 2 (2018), 196–203. doi:10.1109/TCE.2018.2844736

[3] Sai Balaji, Nachum Plonka, Mike Atkinson, Malathy Muthu, Minvydas Ragulskis, Alfonsas Vainoras, and Rollin McCraty. 2025. Heart Rate Variability Biofeedback in a Global Study of the Most Common Coherence Frequencies and the Impact of Emotional States. *Scientific Reports* 15, 1 (2025), 3241. doi:10.1038/s41598-025-87729-7

[4] Lisa Feldman Barrett, Ralph Adolphs, Stacy Marsella, Aleix M. Martinez, and Seth D. Pollak. 2019. Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements. *Psychological Science in the Public Interest* 20, 1 (2019), 1–68. doi:10.1177/1529100619832930

[5] Geoff Cumming. 2014. The New Statistics: Why and How. *Psychological Science* 25, 1 (2014), 7–29. doi:10.1177/0956797613504966

[6] Tuomas Eerola and Jonna K. Vuoskoski. 2013. A Review of Music and Emotion Studies: Approaches, Emotion Models, and Stimuli. *Music Perception* 30, 3 (2013), 307–340. doi:10.1525/mp.2012.30.3.307

[7] Anthony G. Greenwald. 1976. Within-Subjects Designs: To Use or Not to Use? *Psychological Bulletin* 83, 2 (1976), 314–320. doi:10.1037/0033-2909.83.2.314

[8] Patrick Helmholz, Michael Meyer, and Susanne Robra-Bissantz. 2019. Feel the Moosic: Emotion-Based Music Selection and Recommendation. In *Proceedings of the 32nd Bled eConference: Humanizing Technology for a Sustainable Society*. University of Maribor Press, Maribor, Slovenia, 203–221. doi:10.18690/978-961-286-280-0.11

[9] Joris H. Janssen, Egon L. van den Broek, and Joyce H. D. M. Westerink. 2012. Tune in to Your Emotions: A Robust Personalized Affective Music Player. *User Modeling and User-Adapted Interaction* 22 (2012), 255–279. doi:10.1007/s11257-011-9107-7

[10] Patrik N. Juslin and Daniel Västfjäll. 2008. Emotional Responses to Music: The Need to Consider Underlying Mechanisms. *Behavioral and Brain Sciences* 31, 5 (2008), 559–575. doi:10.1017/S0140525X08005293

[11] Marius Kaminskas and Francesco Ricci. 2012. Contextual Music Information Retrieval and Recommendation: State of the Art and Challenges. *Computer Science Review* 6, 2–3 (2012), 89–119. doi:10.1016/j.cosrev.2012.04.002

[12] Bart P. Knijnenburg, Martijn C. Willemsen, Zeno Gantner, Hakan Soncu, and Chris Newell. 2012. Explaining the User Experience of Recommender Systems. *User Modeling and User-Adapted Interaction* 22, 4–5 (2012), 441–504. doi:10.1007/s11257-011-9118-4

[13] Peter Kuppens, Francis Tuerlinckx, Michelle Yik, Peter Koval, Joachim Coosemans, Kevin J. Zeng, and James A. Russell. 2017. The Relation Between Valence and Arousal in Subjective Experience Varies With Personality and Culture. *Journal of Personality* 85, 4 (2017), 530–542. doi:10.1111/jopy.12258

[14] Sylvain Laborde, Emma Mosley, and Julian F. Thayer. 2017. Heart Rate Variability and Cardiac Vagal Tone in Psychophysiological Research—Recommendations for Experiment Planning, Data Analysis, and Data Reporting. *Frontiers in Psychology* 8 (2017), 213. doi:10.3389/fpsyg.2017.00213

[15] Daniël Lakens. 2022. Sample Size Justification. *Collabra: Psychology* 8, 1 (2022), 33267. doi:10.1525/collabra.33267

[16] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. doi:10.48550/arXiv.1906.08172 arXiv:1906.08172 [cs.DC]

[17] maharshipandya. 2022. Spotify Tracks Dataset. Kaggle. https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset.

[18] M. Loretto Muñoz, Arie van Roon, Harriëtte Riese, Chris Thio, Emma Oostenbroek, Iris Westrik, Eco J. C. de Geus, Ron Gansevoort, Joop Lefrandt, Ilja M. Nolte, and Harold Snieder. 2015. Validity of (Ultra-)Short Recordings for Heart Rate Variability Measurements. *PLOS ONE* 10, 9 (2015), e0138921. doi:10.1371/journal.pone.0138921

[19] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. 2018. The Preregistration Revolution. *Proceedings of the National Academy of Sciences* 115, 11 (2018), 2600–2606. doi:10.1073/pnas.1708274114

[20] Ingrid Nunes and Dietmar Jannach. 2017. A Systematic Review and Taxonomy of Explanations in Decision Support and Recommender Systems. *User Modeling and User-Adapted Interaction* 27, 3–5 (2017), 393–444. doi:10.1007/s11257-017-9195-0

[21] Jonathan Posner, James A. Russell, and Bradley S. Peterson. 2005. The Circumplex Model of Affect: An Integrative Approach to Affective Neuroscience, Cognitive Development, and Psychopathology. *Development and Psychopathology* 17, 3 (2005), 715–734. doi:10.1017/S0954579405050340

[22] Karen S. Quigley, Peter J. Gianaros, Greg J. Norman, J. Richard Jennings, Gary G. Berntson, and Eco J. C. de Geus. 2024. Publication Guidelines for Human Heart Rate and Heart Rate Variability Studies in Psychophysiology—Part 1: Physiological Underpinnings and Foundations of Measurement. *Psychophysiology* 61, 9 (2024), e14604. doi:10.1111/psyp.14604

[23] James A. Russell. 1980. A Circumplex Model of Affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161–1178. doi:10.1037/h0077714

[24] Thomas Schäfer, Peter Sedlmeier, Christine Städtler, and David Huron. 2013. The Psychological Functions of Music Listening. *Frontiers in Psychology* 4 (2013), 511. doi:10.3389/fpsyg.2013.00511

[25] Markus Schedl, Hamed Zamani, Ching-Wei Chen, Yashar Deldjoo, and Mehdi Elahi. 2018. Current Challenges and Visions in Music Recommender Systems Research. *International Journal of Multimedia Information Retrieval* 7 (2018), 95–116. doi:10.1007/s13735-018-0154-2

[26] Fred Shaffer and J. P. Ginsberg. 2017. An Overview of Heart Rate Variability Metrics and Norms. *Frontiers in Public Health* 5 (2017), 258. doi:10.3389/fpubh.2017.00258

[27] Robert E. Thayer. 1991. *The Biopsychology of Mood and Arousal*. Oxford University Press, New York, NY.

[28] Hieu Tran, Tuan Le, Anh Do, Tram Vu, Steven Bogaerts, and Brian Howard. 2023. Emotion-Aware Music Recommendation. *Proceedings of the AAAI Conference on Artificial Intelligence* 37, 13 (2023), 16087–16095. doi:10.1609/aaai.v37i13.26911

[29] Yi-Hsuan Yang and Homer H. Chen. 2012. Machine Recognition of Music Emotion: A Review. *ACM Transactions on Intelligent Systems and Technology* 3, 3, Article 40 (2012), 30 pages. doi:10.1145/2168752.2168754